4.4 ARMAX Method: Aggregate-Daily (Store Type A)

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Section Overview

Goal By building and comparing various ARMAX models for aggregated daily sales of Store Type A, we wish to answer the following question by the end of this section:

Can we improve sales forecasts by using external variables, and which predictor combination works best?

This notebook:

1. Reads aggregate processed files directly.
2. Uses a single train/holdout split.
3. Selects the best ARMAX specification by train AIC.
4. Evaluates the selected model on holdout using fixed-coefficient 1-step forecasts.
5. Produces aggregate-daily test forecasts.

Summary of Approach:

4.4.1 Define forecasting assumptions and available information 4.4.2 Specify candidate ARMAX models with various specifications 4.4.3 Split data into training and holdout sets (merge wrangling later?) 4.4.4 Explore variable relationships using CCF diagnostics 4.4.5 Fit ARMAX models and select the best specification 4.4.6 Evaluate forecasting performance on the holdout set 4.4.7 Perform residual diagnostics 4.4.8 Refit the best model on full training data and forecast the test period

4.4.0 Data Preparation and Feature Construction

Shall we move library set ups to the beginning of the report later?

```r
library(tidyverse)
library(forecast)

seasonal_frequency <- 7L

train_data <- readr::read_csv(
  "../../data/processed/store_type_a_train_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date),
    PromoShare = dplyr::if_else(OpenStores > 0, PromoStores / OpenStores, 0),
    DayOfWeek = as.integer(DayOfWeek),
    Month = as.integer(Month)
  ) |>
  dplyr::arrange(Date)
```

```r
test_data <- readr::read_csv(
  "../../data/processed/store_type_a_test_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date),
    PromoShare = dplyr::if_else(OpenStores > 0, PromoStores / OpenStores, 0),
    DayOfWeek = as.integer(DayOfWeek),
    Month = as.integer(Month)
  ) |>
  dplyr::arrange(Date)

head(train_data)
```

```
## # A tibble: 6 x 11
##   Date         Sales OpenStores PromoStores PromoShare SchoolHolidayShare
##   <date>       <dbl>      <dbl>       <dbl>      <dbl>              <dbl>
## 1 2013-01-01    2907          1           0          0              1
## 2 2013-01-02 3920386        602           0          0              1
## 3 2013-01-03 3528027        601           0          0              0.939
## 4 2013-01-04 3610215        601           0          0              0.939
## 5 2013-01-05 2964587        600           0          0              0.0664
## 6 2013-01-06    6254          3           0          0              0.0664
## # i 5 more variables: DayOfWeek <int>, Year <dbl>, Month <int>, IsoWeek <dbl>,
## #   IsWeekend <dbl>
```

Interpretation When conducting ARMAX / ARIMAX analysis, we are interested in evaluating exogenous variables that may influence sales (target variable). Here we have:

- `PromoShare`: captures promotional intensity across stores
- `OpenStores`: reflects operational scale
- `DayOfWeek`, `Month`: capture calendar-driven seasonality

These variables form the exogenous variables to be used in the ARMAX model. The goal is to explain variation in sales beyond what can be captured by past values alone.

4.4.1 Forecasting Assumption and Information Set

This ARMAX setup is a conditional forecast: we assume day-t inputs are known when forecasting day t.

For a 1-step-ahead forecast at day t, the model uses:

$$\mathcal{F}_{t-1} \cup \{X_t^{planned}\}$$

Here, X_t includes `PromoShare`, `OpenStores`, and calendar covariates.

Practical interpretation:

1. `DayOfWeek` and `Month` are deterministic and known in advance.

2. `PromoShare` and `OpenStores` are valid only if planned and available by the end of day $t-1$.
3. If you use realized day-t values, holdout RMSE may be overly optimistic due to information leakage beyond $\mathcal{F}_{t-1}$.

All makes great sense - do we need to prove this is the case in the actual dataset when we practice actual analysis?

4.4.2 Candidate ARMAX Model Specifications

We define the following ARMAX specifications with increasing complexity:

- Basic model: operational variables only
- Extended models: include calendar effects

This allows us to assess whether adding additional predictors improves model fit and forecasting performance. Note that while more complex models may fit better, we risk overfitting so complex models must be validated on the holdout set (section 4.4.8).

```
model_specs <- list(
  ARMAX_Promo_Open = stats::as.formula("~ PromoShare + OpenStores"),
  ARMAX_Promo_Open_DOW = stats::as.formula("~ PromoShare + OpenStores + DayOfWeek"),
  ARMAX_Promo_Open_DOW_Month = stats::as.formula("~ PromoShare + OpenStores + DayOfWeek + Month")
)

model_specs
```

```
## $ARMAX_Promo_Open
## ~PromoShare + OpenStores
##
## $ARMAX_Promo_Open_DOW
## ~PromoShare + OpenStores + DayOfWeek
##
## $ARMAX_Promo_Open_DOW_Month
## ~PromoShare + OpenStores + DayOfWeek + Month
```

4.4.3 Holdout Split Setup

potentially merge with wrangling for final version

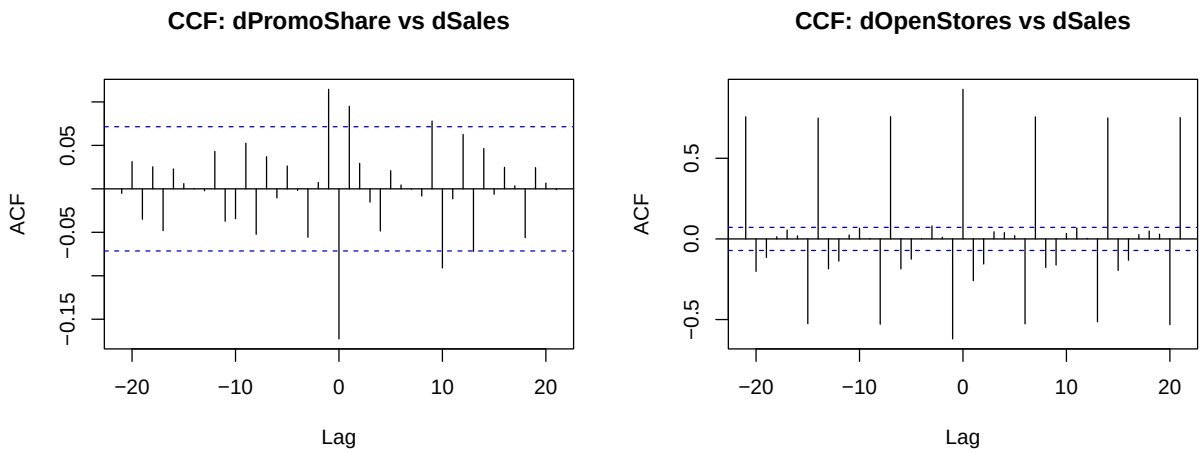
```
set.seed(443)

holdout_prop <- 0.20
n_total <- nrow(train_data)
n_train <- floor((1 - holdout_prop) * n_total)

train_subset <- train_data |> dplyr::slice(1:n_train)
holdout_subset <- train_data |> dplyr::slice((n_train + 1):n_total)

holdout_split <- tibble::tibble(
  n_total = n_total,
```

```r
  n_train = n_train,
  n_holdout = n_total - n_train,
  train_start = min(train_subset$Date),
  train_end = max(train_subset$Date),
  holdout_start = min(holdout_subset$Date),
  holdout_end = max(holdout_subset$Date)
)

holdout_split
```

```
## # A tibble: 1 x 7
##   n_total n_train n_holdout train_start train_end  holdout_start holdout_end
##     <int>   <dbl>     <dbl> <date>      <date>     <date>        <date>
## 1     942     753       189 2013-01-01  2015-01-23 2015-01-24    2015-07-31
```

4.4.4 Diagnostic Analysis: Cross-Correlation (CCF)

```r
diag_data <- train_subset |>
  dplyr::transmute(
    Sales,
    PromoShare,
    OpenStores
  ) |>
  dplyr::mutate(
    dSales = c(NA_real_, diff(Sales)),
    dPromoShare = c(NA_real_, diff(PromoShare)),
    dOpenStores = c(NA_real_, diff(OpenStores))
  ) |>
  tidyr::drop_na()

old_par <- par(no.readonly = TRUE)
par(mfrow = c(1, 2))

stats::ccf(
  diag_data$dPromoShare,
  diag_data$dSales,
  lag.max = 21,
  main = "CCF: dPromoShare vs dSales"
)

stats::ccf(
  diag_data$dOpenStores,
  diag_data$dSales,
  lag.max = 21,
  main = "CCF: dOpenStores vs dSales"
)
```




```
par(old_par)
```

Interpretation The cross-correlation function (CCF) examines the relationship between changes in predictors and changes in sales.

- Significant correlations at lag 0 suggest contemporaneous relationships.
- Significant correlations at lag k suggest lagged effects between to be completed.

On a high level, these diagnostics provide evidence for including specific predictors in the ARMAX model (ones that show significant correlations with changes in the target variable). Nonetheless, CCF results should be interpreted cautiously as correlation does not imply causation and may be influenced by shared trends or seasonality.

4.4.5 ARMAX Modeling: Function Setup, Fitting and Selection

Setting up ARMAX Modeling Functions The utility functions below standardize the ARMAX modeling process by carrying out the following functionalities:

- Constructing and cleaning exogenous regressors (`xreg`),
- Fitting ARMAX models consistently across specifications,
- Generating fixed-coefficient forecasts for fair evaluation.

This modular design ensures reproducibility and allows systematic comparison across multiple candidate models.

```
build_xreg <- function(df, model_formula) {
  df <- df |>
    dplyr::mutate(
      DayOfWeek = factor(DayOfWeek, levels = 1:7),
      Month = factor(Month, levels = 1:12)
    )
```

```r
  mm <- stats::model.matrix(model_formula, data = df)
  mm[, colnames(mm) != "(Intercept)", drop = FALSE]
}

rmse_vec <- function(actual, pred) {
  sqrt(mean((actual - pred) ^ 2, na.rm = TRUE))
}

clean_xreg <- function(xreg_train, xreg_valid) {
  xreg_train <- as.matrix(xreg_train)
  xreg_valid <- as.matrix(xreg_valid)

  keep_non_constant <- apply(xreg_train, 2, function(x) stats::var(x, na.rm = TRUE) > 0)
  xreg_train <- xreg_train[, keep_non_constant, drop = FALSE]
  xreg_valid <- xreg_valid[, keep_non_constant, drop = FALSE]

  qr_fit <- qr(xreg_train)
  keep_idx <- sort(qr_fit$pivot[seq_len(qr_fit$rank)])
  xreg_train <- xreg_train[, keep_idx, drop = FALSE]
  xreg_valid <- xreg_valid[, keep_idx, drop = FALSE]

  list(train = xreg_train, valid = xreg_valid)
}

prepare_xreg_for_split <- function(data, model_formula, n_train) {
  xreg_full_raw <- build_xreg(data, model_formula)
  xreg_train_raw <- xreg_full_raw[seq_len(n_train), , drop = FALSE]

  xreg_clean <- clean_xreg(xreg_train_raw, xreg_full_raw)

  list(
    train = xreg_clean$train,
    full = xreg_clean$valid
  )
}

fit_armax_train <- function(data, model_formula, n_train) {
  y_train <- ts(data$Sales[seq_len(n_train)], frequency = seasonal_frequency)
  xreg_obj <- prepare_xreg_for_split(data, model_formula, n_train)

  fit <- forecast::auto.arima(
    y_train,
    xreg = xreg_obj$train,
    seasonal = TRUE,
    allowmean = FALSE,
    allowdrift = FALSE,
    stepwise = TRUE,
    approximation = TRUE
  )

  list(
    fit = fit,
    xreg_full = xreg_obj$full
```

```r
  )
}

resolve_seasonal_period <- function(ord, y_ts, fallback_period = 7L) {
  m_raw <- suppressWarnings(as.integer(unname(ord["m"])))
  freq_raw <- suppressWarnings(as.integer(stats::frequency(y_ts)))

  if (!is.na(m_raw) && m_raw > 1L) {
    return(m_raw)
  }

  if (!is.na(freq_raw) && freq_raw > 1L) {
    return(freq_raw)
  }

  fallback_period
}

extract_arima_spec <- function(train_fit_obj, y_ts) {
  ord <- forecast::arimaorder(train_fit_obj)

  list(
    ord = ord,
    order = c(
      as.integer(unname(ord["p"])),
      as.integer(unname(ord["d"])),
      as.integer(unname(ord["q"]))
    ),
    seasonal_order = c(
      as.integer(unname(ord["P"])),
      as.integer(unname(ord["D"])),
      as.integer(unname(ord["Q"]))
    ),
    seasonal_period_raw = unname(ord["m"]),
    seasonal_period_used = resolve_seasonal_period(ord, y_ts)
  )
}

fixed_coef_holdout_forecast <- function(data, n_train, train_fit_obj, xreg_full) {
  y_all <- ts(data$Sales, frequency = seasonal_frequency)
  spec <- extract_arima_spec(train_fit_obj, y_all)

  fixed_fit <- stats::arima(
    y_all,
    order = spec$order,
    seasonal = list(
      order = spec$seasonal_order,
      period = spec$seasonal_period_used
    ),
    xreg = xreg_full,
    include.mean = FALSE,
    fixed = as.numeric(train_fit_obj$coef),
    transform.pars = FALSE,
```

```
    optim.control = list(maxit = 0),
    method = "ML"
  )

  fc_all <- as.numeric(y_all - fixed_fit$residuals)
  holdout_idx <- (n_train + 1):length(y_all)

  tibble::tibble(
    Date = data$Date[holdout_idx],
    actual = as.numeric(y_all[holdout_idx]),
    predicted = fc_all[holdout_idx]
  )
}
```

Select ARMAX Specification by Train AIC

Yolanda may add RMSE here in a bit (and recite the numbers in the RMSE section for overall comparison) - depends on where the RMSE calculations make most sense

```
y_train_for_period <- ts(train_data$Sales[seq_len(n_train)], frequency = seasonal_frequency)

train_fit_registry <- purrr::imap(model_specs, function(model_formula, model_name) {
  fit_armax_train(
    data = train_data,
    model_formula = model_formula,
    n_train = n_train
  )
})

aic_table <- purrr::imap_dfr(train_fit_registry, function(obj, model_name) {
  spec <- extract_arima_spec(obj$fit, y_train_for_period)
  seasonal_period_label <- as.character(as.integer(spec$seasonal_period_used))

  tibble::tibble(
    model = model_name,
    train_aic = stats::AIC(obj$fit),
    seasonal_period_raw = spec$seasonal_period_raw,
    seasonal_period_used = spec$seasonal_period_used,
    arima_order = sprintf(
      "(%d,%d,%d)(%d,%d,%d)[%s]",
      spec$order[1],
      spec$order[2],
      spec$order[3],
      spec$seasonal_order[1],
      spec$seasonal_order[2],
      spec$seasonal_order[3],
      seasonal_period_label
    )
  )
}) |>
  dplyr::arrange(train_aic)

aic_table
```

```
## # A tibble: 3 x 5
##   model           train_aic seasonal_period_raw seasonal_period_used arima_order
##   <chr>               <dbl>               <int>                <int> <chr>
## 1 ARMAX_Promo_Op~    21906.                  NA                    7 (2,0,2)(2,~
## 2 ARMAX_Promo_Op~    21921.                  NA                    7 (2,0,5)(2,~
## 3 ARMAX_Promo_Op~    22073.                  NA                    7 (2,0,2)(2,~
```

```
seasonal_period_issues <- aic_table |>
  dplyr::filter(is.na(seasonal_period_raw))

if (nrow(seasonal_period_issues) > 0) {
  warning("At least one ARMAX candidate reports NA seasonal period from arimaorder(). Using seasonal_pe
}

seasonal_period_issues
```

```
## # A tibble: 3 x 5
##   model           train_aic seasonal_period_raw seasonal_period_used arima_order
##   <chr>               <dbl>               <int>                <int> <chr>
## 1 ARMAX_Promo_Op~    21906.                  NA                    7 (2,0,2)(2,~
## 2 ARMAX_Promo_Op~    21921.                  NA                    7 (2,0,5)(2,~
## 3 ARMAX_Promo_Op~    22073.                  NA                    7 (2,0,2)(2,~
```

Interpretation In the code above we fit each candidate ARMAX model on the training set and compare them using AIC.

- A lower AIC indicates a better balance between model fit and complexity.
- This step serves as an initial filter for selecting promising models.

However, note that AIC is an in-sample metric and does not directly measure forecasting performance. Therefore, the selected model must be validated using the holdout dataset.

4.4.6 Fixed-Coefficient 1-Step Holdout Evaluation

In this step we evaluate a model's forecasting performance using 1-step-ahead predictions on the holdout set, where model parameters are fixed from the training stage. This avoids re-fitting and ensures a fair evaluation.

In addition to AIC, RMSE is also used here as a performance metric, where lower RMSE indicates better predictive accuracy. Generally speaking, RMSE is the primary criterion for model comparison in forecasting contexts. need to add rmse

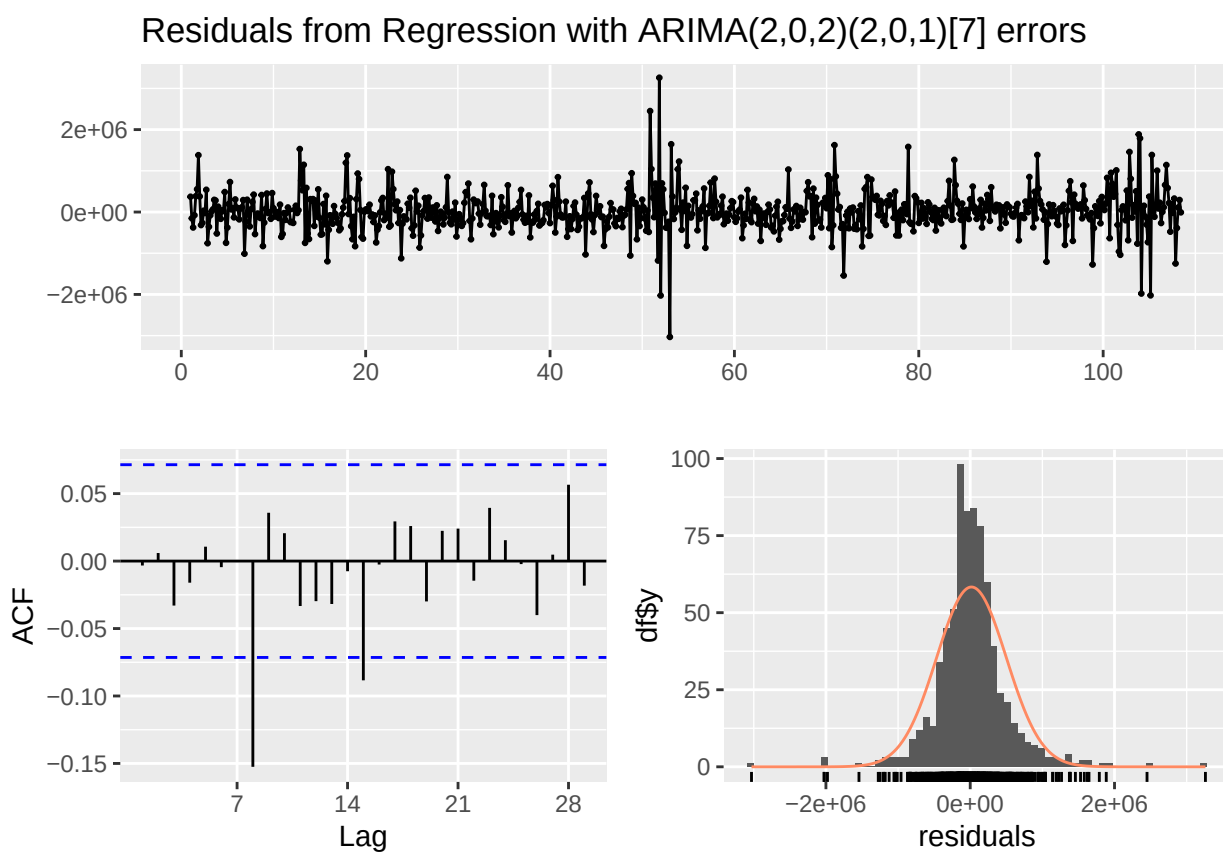
```
best_model <- aic_table |>
  dplyr::slice_min(order_by = train_aic, n = 1)

best_model_name <- best_model$model[[1]]
best_model_formula <- model_specs[[best_model_name]]
best_train_fit <- train_fit_registry[[best_model_name]]$fit
best_xreg_full <- train_fit_registry[[best_model_name]]$xreg_full
```

```r
holdout_predictions <- fixed_coef_holdout_forecast(
  data = train_data,
  n_train = n_train,
  train_fit_obj = best_train_fit,
  xreg_full = best_xreg_full
)

armax_holdout_summary <- tibble::tibble(
  model = best_model_name,
  train_aic = best_model$train_aic[[1]],
  holdout_rmse = rmse_vec(holdout_predictions$actual, holdout_predictions$predicted),
  n_holdout = nrow(holdout_predictions)
)

armax_holdout_summary
```

```
## # A tibble: 1 x 4
##   model                       train_aic holdout_rmse n_holdout
##   <chr>                           <dbl>        <dbl>     <int>
## 1 ARMAX_Promo_Open_DOW_Month    21906.      399948.       189
```

4.4.7 Residual Diagnostics

```r
forecast::checkresiduals(best_train_fit)
```

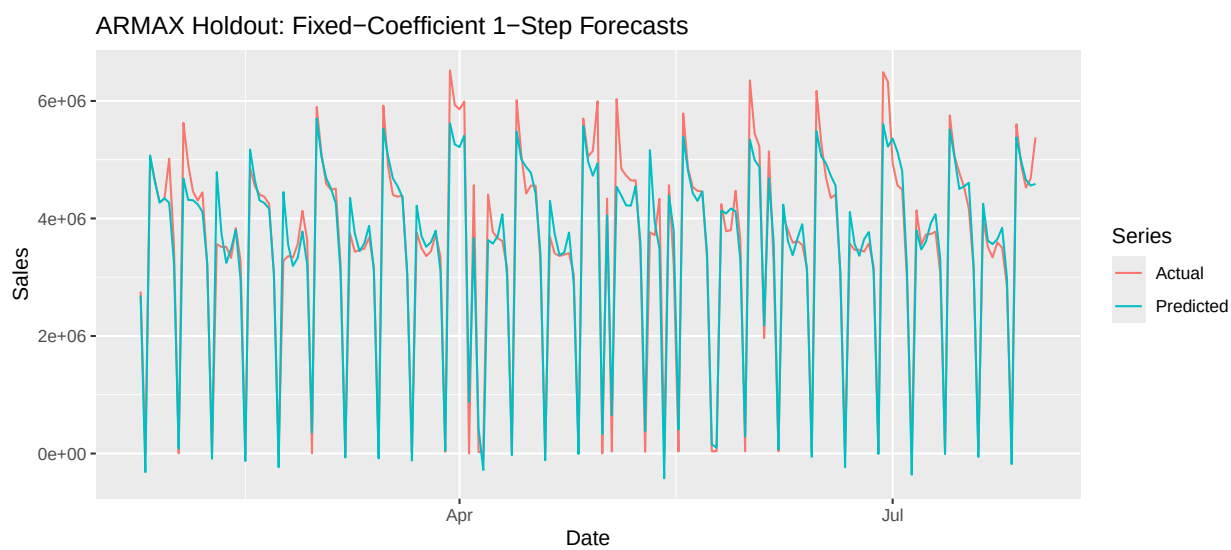


```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(2,0,2)(2,0,1)[7] errors
## Q* = 22.552, df = 7, p-value = 0.002039
##
## Model df: 7.   Total lags used: 14
```

```r
residual_diagnostics <- tibble::tibble(
  model = best_model_name,
  sigma2 = best_train_fit$sigma2,
  ljung_box_pvalue = stats::Box.test(
    residuals(best_train_fit),
    lag = 14,
    type = "Ljung-Box",
    fitdf = sum(best_train_fit$arma[1:4])
  )$p.value
)

residual_diagnostics
```

```
## # A tibble: 1 x 3
##   model                              sigma2 ljung_box_pvalue
##   <chr>                               <dbl>            <dbl>
## 1 ARMAX_Promo_Open_DOW_Month 242257433257.          0.00204
```

```r
ggplot(holdout_predictions, aes(x = Date)) +
  geom_line(aes(y = actual, color = "Actual")) +
  geom_line(aes(y = predicted, color = "Predicted")) +
  labs(
    title = "ARMAX Holdout: Fixed-Coefficient 1-Step Forecasts",
    x = "Date",
    y = "Sales",
    color = "Series"
  )
```



Interpretation

Residual diagnostics assess whether the model has captured the underlying time series structure.

- Ideally, residuals should resemble white noise.
- The Ljung–Box test evaluates whether significant autocorrelation remains.

If residuals are not white noise, this suggests that the model may be misspecified and could be improved by adjusting AR or MA components. add comments to tie into actual plots - yolanda

Best-Performing ARMAX Model

do we want to add a version here with the best model variables but weekends removed?

```
best_model
```

```
## # A tibble: 1 x 5
##   model           train_aic seasonal_period_raw seasonal_period_used arima_order
##   <chr>               <dbl>               <int>                <int> <chr>
## 1 ARMAX_Promo_Op~    21906.                  NA                    7 (2,0,2)(2,~
```

```
best_model_name
```

```
## [1] "ARMAX_Promo_Open_DOW_Month"
```

```
best_model_formula
```

```
## ~PromoShare + OpenStores + DayOfWeek + Month
```

The selected model above represents the best-performing ARMAX specification based on the chosen selection criteria.

This model balances: - Inclusion of relevant exogenous variables, as well as
- Adequate modeling of time series dependence.

Hence, it will be the model we proceed with for final forecasting in this section.

4.4.8 Test Forecasting with Final Model

In this step, the selected ARMAX model is refit using the full training dataset to maximize information usage. Forecasts are then generated for the test period using the corresponding exogenous variables. These forecasts represent the final model output and will be used for evaluation and comparison with other methods.

```
y_full <- ts(train_data$Sales, frequency = seasonal_frequency)

xreg_train_full_raw <- build_xreg(train_data, best_model_formula)
xreg_test_raw <- build_xreg(test_data, best_model_formula)

xreg_clean_full <- clean_xreg(xreg_train_full_raw, xreg_test_raw)
xreg_train_full <- xreg_clean_full$train
xreg_test <- xreg_clean_full$valid
```

```r
best_fit <- forecast::auto.arima(
  y_full,
  xreg = xreg_train_full,
  seasonal = TRUE,
  allowmean = FALSE,
  allowdrift = FALSE,
  stepwise = TRUE,
  approximation = TRUE
)
test_forecast <- as.numeric(forecast::forecast(best_fit, xreg = xreg_test, h = nrow(test_data))$mean)

aggregate_test_forecast <- test_data |>
  dplyr::transmute(
    Date,
    predicted_aggregate_sales = pmax(0, test_forecast)
  )

head(aggregate_test_forecast)
```

```
## # A tibble: 6 x 2
##   Date       predicted_aggregate_sales
##   <date>                         <dbl>
## 1 2015-08-01                  2515834.
## 2 2015-08-02                   288768.
## 3 2015-08-03                  3487401.
## 4 2015-08-04                  2847204.
## 5 2015-08-05                  2554873.
## 6 2015-08-06                  2526256.
```

Save Outputs

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(holdout_split, "results/armax_holdout_split.csv")
readr::write_csv(aic_table, "results/armax_aic_table.csv")
readr::write_csv(holdout_predictions, "results/armax_holdout_predictions.csv")
readr::write_csv(armax_holdout_summary, "results/armax_holdout_rmse_summary.csv")
readr::write_csv(aggregate_test_forecast, "results/armax_aggregate_test_forecast.csv")
```